

IoT Homework 2025/2026

Integrated Report: BLE Localization

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Exercise 2 - BLE Localization of Dental Prostheses

The original firmware sends one BLE advertisement every 10 s for the whole day. This produces 8640 advertisements/day. The objective is to preserve localization when it is useful while reducing unnecessary BLE transmissions.

2.1 Beaconing logic

The proposed redesign is a state-based beaconing policy driven by a worn detector and a motion detector:

State	Condition	BLE behavior	Reason
WORN	Denture detected in the mouth	Rare heartbeat every 5 min, or BLE fully off in strict mode	The prosthesis is with the patient, so continuous localization is not useful
NOT_WORN_STILL	Outside the mouth and no motion	Slow advertising every 30 s	The position is not changing; slow RSSI updates are enough
NOT_WORN_MOVING	Outside the mouth and motion/search detected	Fast advertising every 10 s	Preserves the original localization update rate when movement matters

2.2 Added sensors

The additional sensing architecture contains:

- In-mouth / worn detector: capacitive contact or proximity sensing through sealed electrodes. A temperature check can be used as confirmation to avoid false positives.
- Motion detector: ultra-low-power accelerometer with wake-on-motion interrupt, used only to switch between still and moving localization modes.
- BLE SoC: a low-power microcontroller/radio that can remain in deep sleep and wake on sensor interrupts.

2.3 Improved firmware pseudocode

```
# Constants
MOVING_INTERVAL = 10 seconds      # original localization update rate
STILL_INTERVAL = 30 seconds       # slow keep-alive when not worn and still
WORN_HEARTBEAT_INTERVAL = 300 seconds # robust mode; strict mode can disable BLE
MOTION_TIMEOUT = 120 seconds

# State variable
last_motion_time = 0

function on_motion_interrupt():
    last_motion_time = current_time()

function app_main():
```

```

setup_BLE()
setup_in_mouth_sensor()
setup_accelerometer_interrupt(callback = on_motion_interrupt)

loop forever:
    worn = read_in_mouth_sensor()

    if worn == TRUE:
        # Denture is inside the patient's mouth.
        # Continuous localization is not needed.
        send_BLE_beacon(state = WORN, battery_level)
        sleep(WORN_HEARTBEAT_INTERVAL)
        continue

    # Denture is outside the mouth and must remain localizable.
    if motion_detected() == TRUE:
        last_motion_time = current_time()

    if current_time() - last_motion_time < MOTION_TIMEOUT:
        send_BLE_beacon(state = NOT_WORN_MOVING, battery_level)
        sleep(MOVING_INTERVAL)
    else:
        send_BLE_beacon(state = NOT_WORN_STILL, battery_level)
        sleep(STILL_INTERVAL)

```

2.4 Commercial components

Function	Example commercial part	Role in the design
BLE SoC	Nordic Semiconductor nRF52832 / nRF52810	BLE advertising, sensor interface, low-power sleep
In-mouth capacitive sensing	Azoteq IQS211A or TI FDC1004	Detects saliva/skin contact through sealed electrodes
Temperature confirmation	Murata NCP15XH103 NTC or TI TMP117	Optional confirmation of near-body temperature
Motion detection	Analog Devices ADXL362	Wake-on-motion interrupt and moving/still classification
Battery	CR2032 coin cell	Small form-factor energy source for denture beacon

The electronics must be sealed against saliva and cleaning liquids. Any electrode or surface in contact with the mouth must be biocompatible and mechanically protected.

2.5 Energy-saving estimate

The estimate focuses on BLE advertising events, because the same BLE advertisement energy is used in the old and new firmware. The absolute total energy depends on the exact BLE chip configuration and sensor duty cycle.

Assumptions:

- Original firmware: one advertisement every 10 s for 24 h/day.
- Proposed robust mode: worn 16 h/day with one heartbeat every 300 s.
- Not worn and stationary: 7 h/day with one advertisement every 30 s.
- Not worn and moving/searching: 1 h/day with one advertisement every 10 s.
- Illustrative advertisement energy: $V = 3\text{ V}$, $I_{adv} = 8\text{ mA}$, $t_{adv} = 3\text{ ms}$, so $E_{adv} = 72\text{ microjoules}$.

Case	Beacon count/day	Advertising energy/day
Original periodic firmware	8640	622.1 mJ
WORN heartbeat	192	13.8 mJ
NOT_WORN_STILL	840	60.5 mJ
NOT_WORN_MOVING	360	25.9 mJ
Proposed total	1392	100.2 mJ

Therefore, the BLE-advertising traffic saving is $1 - 1392/8640 = 0.839$, i.e. approximately 83.9%. The same percentage applies to the advertising energy under the equal-Eadv assumption. If the strict mode is used with BLE fully disabled while worn, the daily beacon count decreases further to 1200, giving about 86.1% advertising-energy saving.

Final Numerical Values for Form Entry

Quantity	Value
Exercise 2 - original beacons/day	8640
Exercise 2 - proposed beacons/day	1392
Exercise 2 - advertising saving	83.9%